PRACTICAL NO. 4

WELDING SHOP

INTRODUCTION:

Welding is a process of joining similar metals by application of heat with or without application of pressure and addition of filler material. The results is a continuity of homogeneous material, of the composition and characteristics of two parts which are being joined together. Welding is extensively used in the fabrication work in which metal plates, rolled steel sections, castings of ferrous materials are joined together it is also used for repairing broken, worn-out or defective metal parts.

TYPES OF WELDING

There are two main types of welding they are:

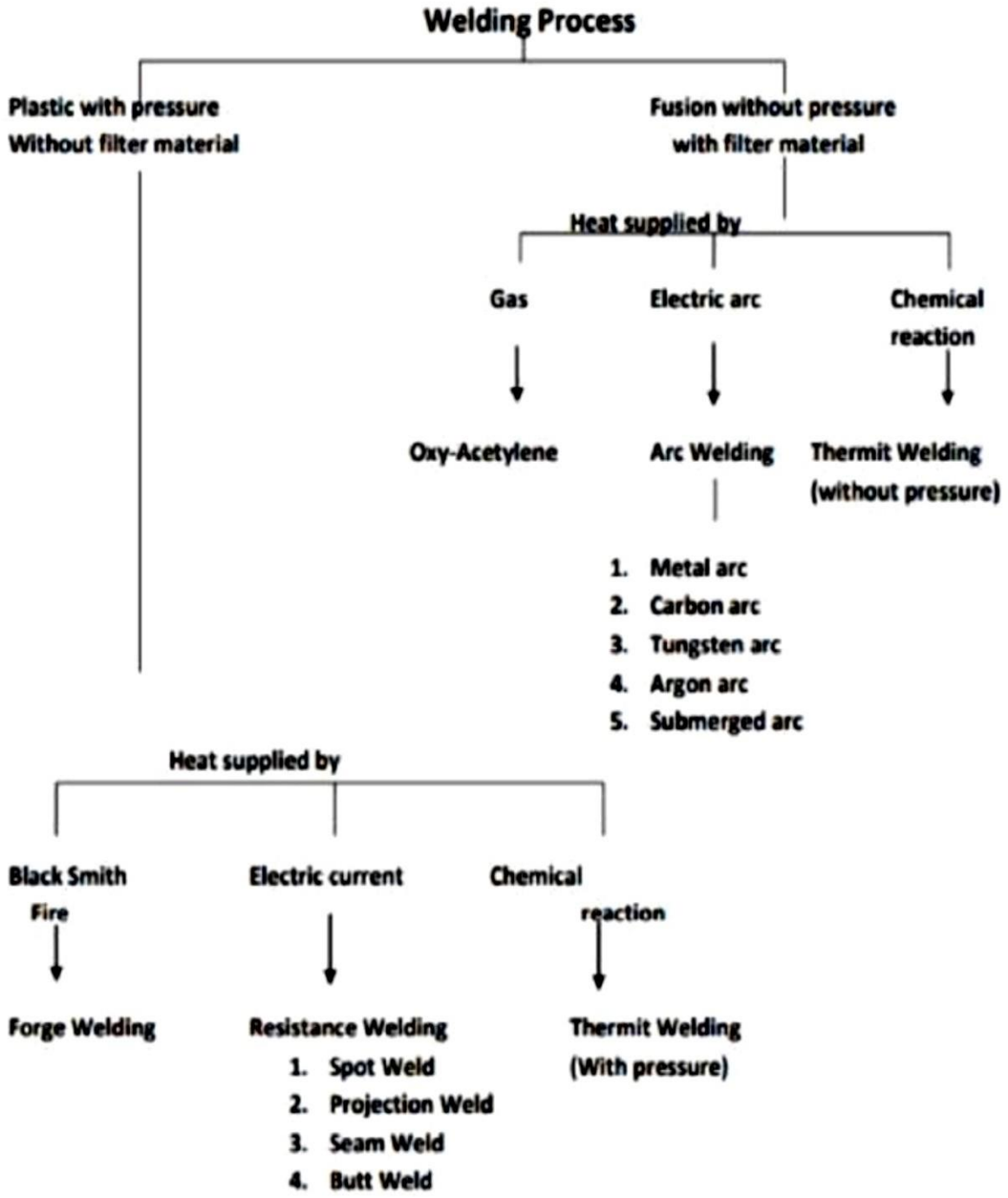
1. Plastic (Pressure) welding

2. Fusion (Non-Pressure) welding

In the plastic welding or pressure welding, the pieces of metal to be joined are heated to a plastic state and then forced together by external pressure. This procedure is used in forge welding, resistance welding, “thermit” welding, and gas welding, in which: pressure is required.

In the fusion welding or non-pressure welding, the material at the joint is heated to a molten state and allowed to solidify. This includes gas welding, arc welding “thermit" welding, etc.

CLASSIFICATION OF WELDING



1. ARC WELDING

Arc welding is metal joining process, where the joint is produced by heating the work piece with an electric arc set up between an electrode and the work piece. The electrical energy is converted is into heat in the arc, which attains a temperature around 5500°Celcius, the electrode itself melts and supplies the necessary filler material.

PRINCIPLE OF ARC WELDING

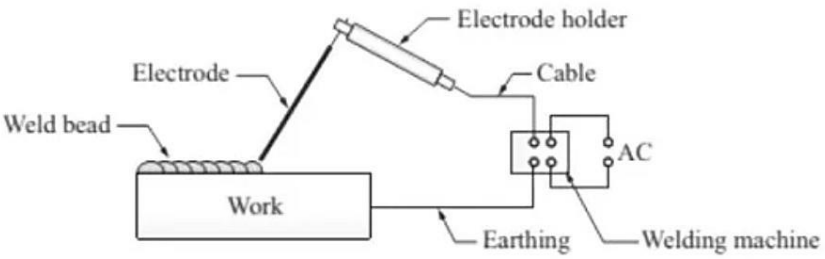


Fig. : Schematic view of arc welding technique

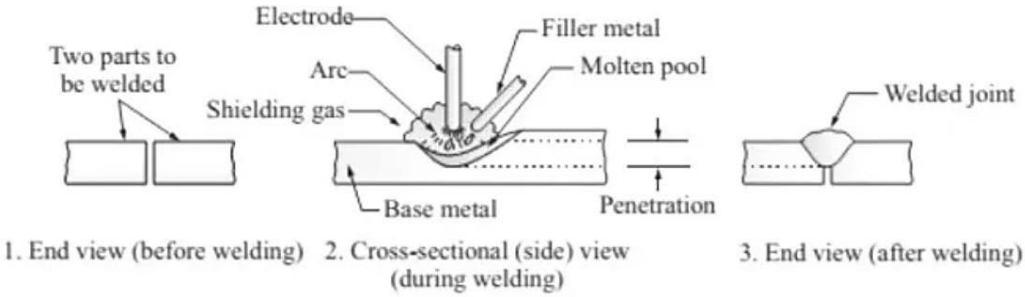


Fig 1: Arc Welding Process

As shown in fig in are welding metal rod is used as one electrode, while the work being welded is used as another electrode during welding operation, this metal electrode is melted by the heat of the arc and is fused with the base metal, thus forming a solid uniform weld, after the metal has been cooled. Both AC and DC may be used for this welding process.

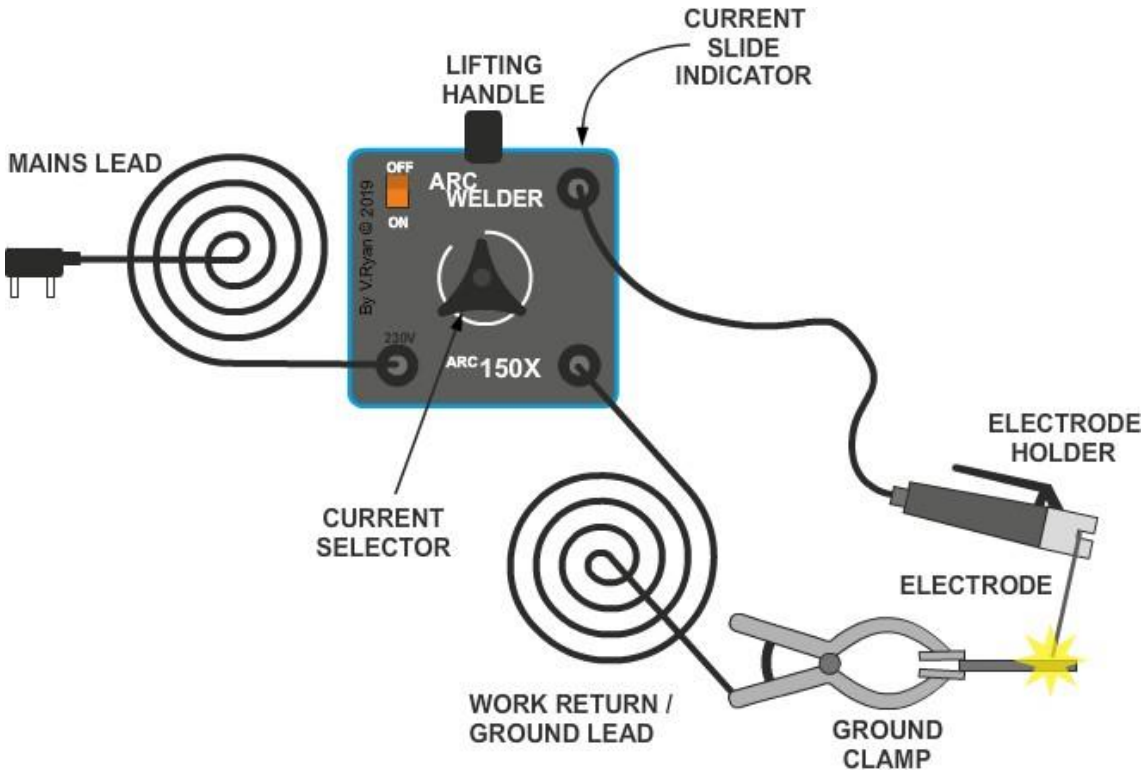
2. ELECTRIC ARC WELDING

The Process is illustrated by means of a schematic diagram the metal of the work piece to be joined is called base metal or parent metal and that provided by the electrode as filler metal. Arc welding is the most extensively employed method of joining metal parts. The source of heat is an electric arc.

The arc column is generated between an anode, which is the positive pole of DC (Direct current) power supply, and the cathode, the negative pole.

When these two conductors of an electric circuit are brought together and separated fora small distance (2 to 4 mm) such that the current continues to flow through a patch Fig. (2) of ionized particles (gaseous medium), called plasma, an electric arc is formed. Heat is generated as the ions strike the cathode. However, electrical energy is converted ta heat energy. Approximately 1 kWh of electricity will create 250 calories (1000 J), the temperature at the center of the arc being 6,000 to 7,000°C. The temperature of an electric are, of course, depends upon the type of electrodes between which it is struck.

The heat of the arc raises the temperature of the parent metal which is melted forming a pool of molten metal. The electrode metal (in metal arc welding) or welding rod (in carbon-arc welding) is also melted and is transferred into the metal in the form of globules of molten metal. The deposited metal serves to fill and bond the joint or to fuse and build up the parent metal surface.



3. ARC WELDING ELECTRODES

There are 2 types of electrodes are used in Arc welding.
1. Consumable electrode
2. Non-consumable electrode

Consumable electrodes melts along with the work piece and fills in the joint. When non-consumable electrodes are used, an additional filler material is used, the advantage using non-consumable-electrode is the metal deposited by a filler rod can be controlled which is not possible in consumable electrode.

4. FLUXES

An electrode has metal core surrounded by flux coating. Flux coating is made up of ferrous alloys. The functions of flux coating are

1. It produces a gas which provides a shield around the arc to protect it from atmosphere.
2. It forms slag by mixing with impurities of the molten metal and thus, refines the metal.
3. The slag, being lighter, floats aver the surface of the molten metal and on solidification forms a thin layer over the weldment, which helps in gradual and uniform cooling of weld and prevents its oxidation during cooling.
4. In some cases, it also carried necessary alloying elements which are added to the molten metal.
5. It promotes conduction of electric current across the arc and helps in stabilizing the arc.
6. It also helps in controlling the bead shape by providing necessary materials for this purpose.

EQUIPMENTS USED IN ARC WELDING

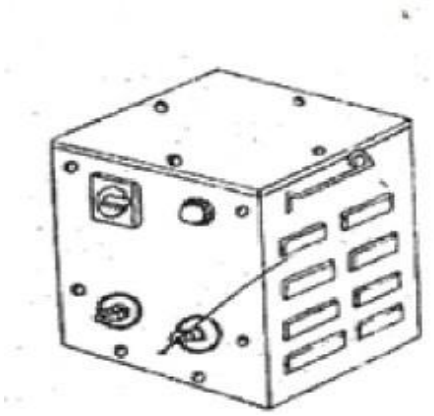
The most commonly used equipment for arc: welding consists of

1. AC or DC machine (transformer or generator)
2. Electrode
3. Electrode holder,
4. Cables, cable connectors
5. Earthing clamps
6. Safety goggles
7. Welding helmet
8. Hand gloves
9. Aprons
10. Chipping hammer
11. Wire brush etc.,

Figure shows the carpenters Bench Vice, used as a work holding device in a carpenter shop. Its one jaw is fi**xe**d to the side of the table while the other jaw is movable by means of a screw and a handle. The jaws are lined with hard wooden faces.

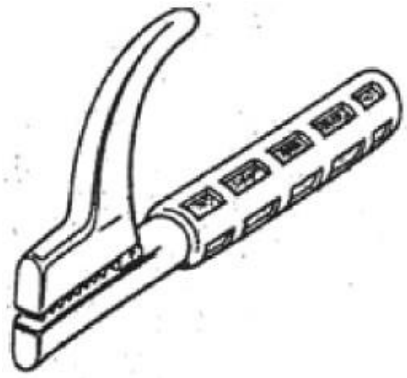
- **TRANSFORMER**

The function of a transformers to generate a low voltage (10 to 50 V), and high amperage (50.4 to 300A) Electric Current. The current may be alternating current or direct current. Transformer converts Electrical energy into heat Energy which is required for the welding process.



- **ELECTRODE HOLDER**

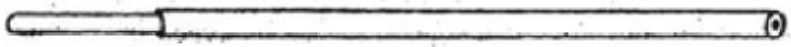
The electrode holder is connected to the end of the welding cable and holds the electrode. It should be light strong and easy to handle and should not become hot while in operation. Adjacent **Fig.** shows one type of electrode holder. The jaws of the holder are insulated, offering protection from electric shock.



- **ELECTRODE**

Welding rod as shown in fig. is one of the electrode which acts as a filler material i.e., when the arc produced at the tip of electrode during welding process, the electrode itself malts and fills the gap between two base metals

Electrodes commonly used are of two types
(i) Coated electrodes (ii) Bare electrode



Coated electrodes carry a core of bare metallic coated wire provided with a flux coating wire provide with a flux coating or covering on the outside surface.

Mild steal is the most commonly used material for core wire.

The common ingredients of a flux which help in slag formation and metal refining are asbestos, mica, silica, fluorspar, stealite etc..

Bare electrodes is one which does not have any coating. Bare electrodes are

cheaper, but welds produced through these are of poor quality. It is used in inert gas metal arc welding (MIG).

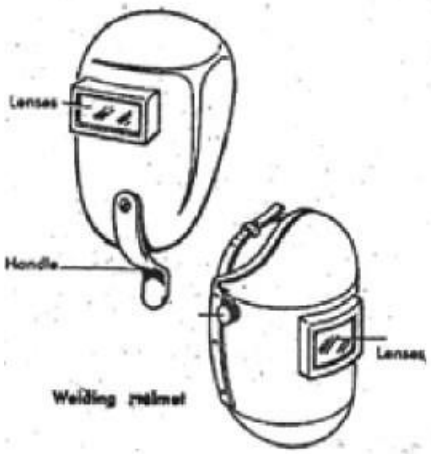
It is used to cut across the grains of the stock. The teeth are so set that the saw kerfs will be wider than the blade thickness. This allows the blade to move freely in tie cut without sticking.

- **GROUND (EARTHIIG) CLAMP**

It is connected to the end of the ground cable and is clamped to the work or welding table to complete the electric circuit as shown in fig. It should be strong and durable, and give low resistance connection.

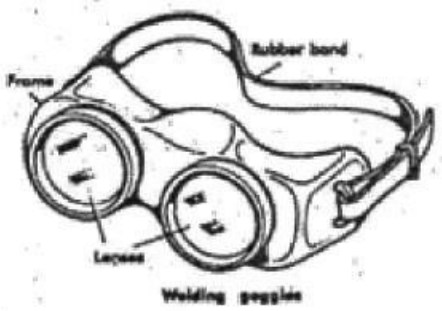
- **FACE OR HEAD SHIELDS**

A face shields provide better protection and allow the welder the free use of both hands. It is mainly used to protect the eyes and face from the rays of the arc and from the spotter or flying particles of hot metal. It is available either in hand or helmet type as shown in fig. The hand type is convenient to use wherever the Work can be done with one hand. The helmet type though not comfortable to wear, leaves both the hands free for the work are made of light weight, non-reflecting fiber fitted with dark glass to filter.



- **GOGGLES**

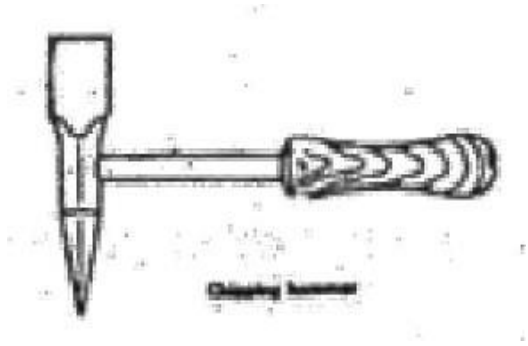
Goggles with coloured glasses are used to protect the eyes from glare and flying bits of hot metal as shown in fig.



<u>Parts of welding goggles: frame, lenses, rubber band</u>

- **CHIPPING HAMMER**

Chipping hammer shown in fig. is used for removing & chipping off the slag that forms on welded surface. A chipping hammer has two striking ends, one end of the head is sharpened like a cold chisel and the other flat end that turns parallel to the handle.



- **WIRE BRUSH**

It has a flat a striking face which is used to give light blows to tools having wooden handle like chisel. It is made of hard wood and is round or rectangular in cross-section.

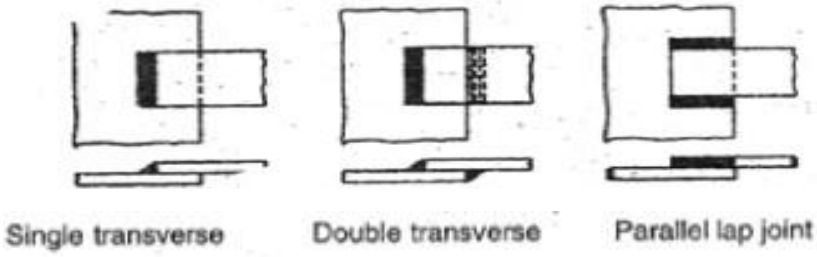
- **APRONS AND HAND GLOVES**

Aprons and hand gloves as show in fig. are flame reardent outfits worm by a welder to protect the under clothing and the body from the sparks the molten metal and the hot metal being welded.

- **TYPES OF WELDED JOINTS**

The relative positions of the two work pieces being joined determine the type of joint. The following are the give basic types of joints commonly used in fusion welding are:

- Lap joint
- Butt joint
- Corner joint
- T-joint
- Edge joint



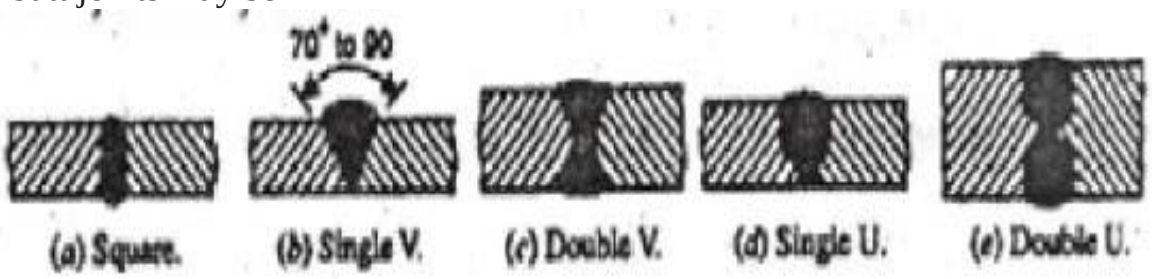
1. Lap joint: The lap joints is obtained by over lapping the plates and then welding the edges of the plates. These joints are employed on plates having thickness less than 3 mm. The lap joints may be:

1. Single transverse
2. Double transverse
3. Parallel lap joint

A shown in fig. , the single transverse lap joints has the disadvantage that the edge-of the plate which is not welded can buckle or warp out of shape.
Differences between Wood File and Metal File.

2. Butt joint

The Butt joint is obtained by welding the ends or edges of the two plates which are approximately in the same plane with each other as shown in fig. In Butt welds, the plate edges do not require beveling if the thickness of plate is less than 5 MM, on the other hand, if the plate thickness is 5 mm to 12.5 mm, the edges should be beveled to V or U - groove and plates having thickness above 12.5 mm should have a V or U - groove on both sides.
The butt joints may be



1. Square butt joint
2. Single V-butt joint
3. Double V-butt joint
4. Single U-butt joint
5. Double U-butt joint

These joints are shown in fig.

3. Corner Joint

The corner joint, as shown in fig. is obtained by joining the edges of two plates whose surfaces are at an angle of approximately 90° to each other. It is used for both light and heavy gauge sheet metal. In some cases corner joint can be welded; without any Fig. filler material, by melting of the edges of the parent metal.

4. Edge joint

The edge joint as shown in fig., is obtained by joining two parallel plates. It is economical for plates having thickness less than 6mm, this joint is unsuitable for members subjected to direct tension or bending.

5. T-Joint

The T-Joint, as shown in fig. is obtained by joining two plates whose surfaces are approximately at night angles to each other.it is widely used to weld stiffeners in air craft and other thin walled structures. These joints are suitable upto 3 mm thickness.

- **COMPARISON BETWEEN A.C. AND D.C ARC WELDING**

Following is the comparison between A.C. and D.C are welding

Use of A.C in arc welding	Use of D.C in are welding
1. An A.C. welding transformer is cheaper and simpler in operation	1. 4.D.C. generator set is costlier and more cumbersome in operation.
2. Maintenance of an A.C. transformer is easier and more economical since it has no moving parts	2. A.D.C. generator carries many moving parts and its maintenance cost is higher.
3. It is less suitable for use at low current with small dia. Electrodes	3. It is better Suited for use at low Amperages with small dia. Electrodes.
4. Except in case of iron powder electrodes, maintenance of a small arc is difficult	4.Maintenance of short arc is easier with D.C.
5. lt is preferred for welding at very large distances from the power supply, because voltage drop in long leads is much less.	5. In D.C. the voltage drop is relatively higher and therefore, only short cables are used, prohibiting its use for welding at long distances from power supply D.C.
6. Striking of arc, particularly with electrodes is relatively difficult.	6. In D.C. it is easier to strike an arc even with thin electrodes.
7. Bare electrodes cannot be used in A.C. only specifically designed coated electrodes with coverings containing stabilizers can be used.	7. Both bare and coated electrodes can be used.

8. Though it can be used for welding positions but selection of proper electrode has to be made carefully and used of a better skill is needed.	8. It is easier to use DLC. even for out of-position welding and for thicker sections because lower currents can be used.
9. It is generally not preferred for welding of sheet metal due to the difficulty in starting the arc.	9. It is more preferred because starting of arc is easier and the arc remains steady.
10. There is hardly any problem of are-blow in A.C.	10. With D.C. there is always a likelihood of arc-blow, unless proper corrective measures are not adopted.
11. Different fixed polarities are not available. Hence it is not suitable for welding all metals, particularly non-ferrous ones.	11. Distinct fixed polarities can be used for welding almost all metals and different thickness.
12. It can be used only when A.C main supply is available	12. An engine driven D.C. generator set can be-used even in absence of A.C. mains supply

- **PREPARATION OF THE WORK BEFORE WELDING :-**

Before welding, the work pieces must be thoroughly cleaned of rust, scale and other foreign materials. Thin pieces of metal are generally weld without beveling the edges. However thick work pieces should be beveled to ensure adequate penetration and fusion of all parts of the weld. But, in either case, the parts to be welded must be separated slightly to allow better penetration of the weld.

- **ADVANTAGES OF WELDING**

1) A properly made weld can be stronger than the part on which it is used.
2) It is a permanent joint.
3) The equipment is inexpensive.
4) The joint produced by welding is as strong as base metal.
5) No patterns are used as in castings.
6) The equipment can be portable.
7) The process allows considerable freedom in design.

- **DISADVANTAGES OF WELDING**

1) A good welding job requires skilled operator.
2) Fixtures are often needed to hold parts in position for welding.
3) Each part of weldment must be cut to size and shape before it can be welded.
4) Presence of residual stresses and distortion in the welded joints.

- **COMMON DEFECTS IN WELDING**

The common defects of welded joints are as follows:

1) Cracks

Cracks may be microscopic scale or macroscopic scale depending upon their size. Cracks in the welded joint may arise from localized stresses set up by uneven heating and cooling, presence of high percentage of sulphur and carbon in the base metal.

2) Porosity

Porosity is a group of small voids, whereas blow holes or gas packet is a comparatively bigger isolated hole or cavity. They occur mainly due to entrapped gasses.

3) Poor fusion

It is lack of thorough and complete union between the metal deposited by the electrodes and parent metal.

4) Inclusions

The presence of non-metallic substances in the metal is called inclusion. Inclusion lowers the strength of the joint.

5) Undercut

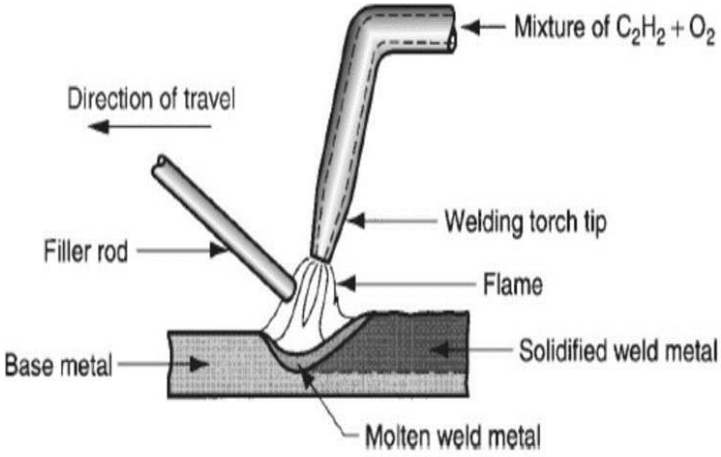
In undercut, a groove gets formed in the base metal along the sides of the welded joint. The reasons for under cutting are non-uniform feed of the welding rod, improper electrode or excessive heating.

- **SAFE WELDING PRACTICES**

1) Never look at the arc with naked eye. The arc can burn your eyes severely. Therefore always use a face or head shield or goggles while welding.
2) Always wear the safety hand gloves apron and shoes.
3) Ensure proper insulation of the cables and check for openings.

- **GAS WELDING**

Gas welding is done by burning a combustible gas with air or oxygen in a concentrated flame of high temperature. As with other welding methods, the purpose of the flame is to heat and melt the parent metal and filler rod of a joint. It can weld most common materials, Equipment

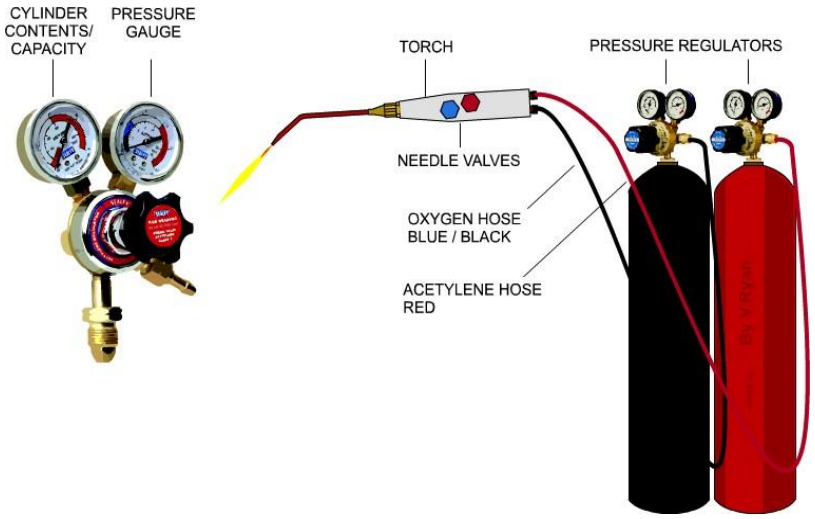


is inexpensive, versatile, and serves adequately in many job and general repair shops.

- **OXY-ACETYLENE WELDING**

Oxy-acetylene gas welding is accomplished by melting the edges or-surface to be joined by gas flame and allowing the molten metal to flaw together, thus forming: a solid continuous joint upon cooling.

This process is particularly suitable for joining metal sheets and plates having thickness of 2 to 50 mm. With material thicker than 15 mm, additional metal calléd filler metal is added to the weld in the form of welding rod. The composition if the filler rod is usually the same or nearly the same as that of the part being welded. To remove the impurities and oxides present on the surfaces of metal to be joined and to obtain a satisfactory bond a flux is always employed during the welding.



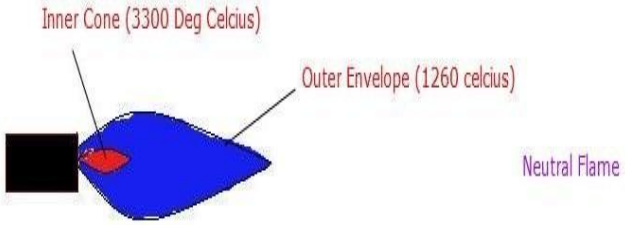
Various gas combination can be used for producing a hot flame for welding metals. Common mixture of gases are oxygen and acetylene, oxygen and hydrogen, oxygen and other fuel gas, and air and acetylene. The oxygen-acetylene mixture is used to. A much greater extent than the other and has a prominent place in the welding industry. The temperature of the oxy-acetylene flame in its hotter region is about 3,200°C.

The oxygen cylinder and acetylene cylinder are connected through regulating valves and pressured into flexible hoses attached to a welding torch. A typical arrangement is shown in **fig.** The welding torch mixes the oxygen and acetylene in the proportions set by the pressure regulators and the final adjustment is done manually at the torch.

The acetylene gas must be handled very carefully as it is explosive at 172 KPa, Therefore, acetylene is stored by dissolving it in acetone which is held in asbestos. In this way it can be stored at 1.4 MPa. Because mixture of acetylene and oxygen are highly explosive, precautions must be taken to avoid mixing the gases improperly. All fitting connecting acetylene gas are left hand treaded, and the hoses are coloured red. For oxygen, the hoses are green or black with right hand threaded fittings. This prevents the making of any improper connections.

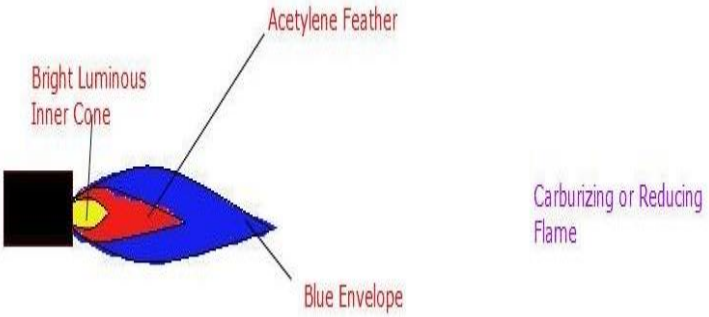
The Correct adjustment of the flame is important for reliable work. When oxygen & acetylene are supplied to torch in different volumes. Three types of flames are produced. They are Neutral flame: when equal volumes of oxygen & acetylene are supplied to torch a neutral flame is produced having a maximum temperature of 3200ºC.

A **neutral flame** has two definite zones: (1) a sharp brilliant cone extending a short distance from the tip of the torch, and (2) an outer cone or envelope only faintly luminous and of a bluish colour. The first one develops heat and the second protects the molten metal from oxidation, because the oxygen in the surrounding atmesphere is consumed by the gases from the flame.

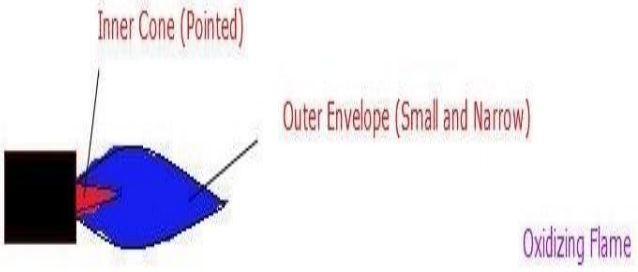


The neutral flame is widely used for welding steel, stainless steel, cast iron, copper, aluminum etc.

A **carburizing flame** is one in which: 'there is an excess of acetylene. This flame has three zones: (1) the sharply defined inner cone (2) an intermediate cone of whitish colour, and (3) the bluish outer cone. The length of the intermediate: cone is an indication of the proportion of excess acetylene in the flame. When welding steel, this will tend to give the steel in the weld a higher carbon content than the parent metal, resulting in a hard and brittle weld.



An **oxidizing flame** is one in which there is an excess of oxygen This flame has two zones (1) the small inner cone which has purplish tinge and (2) the outer cone or envelope. In the case of oxidizing flame the inner cone is not sharply defined as that of neutral or carburizing flame. This flame is necessary for welding brass. In-steel, this will result in a large grain size, increased brittleness with lower strength and elongation.



ADVANTAGES:-

1. The equipment are potable & is comparatively inexpensive & requires little maintenance.
2. Practically all metals may be welded and the equipment can be used for cutting.

DISADVANTAGES:-

1. Gas welding is slower than electric arc welding.
2. Not suitable for heavy section.
3. Gas cylinder should be handled carefully since they are highly explosive.

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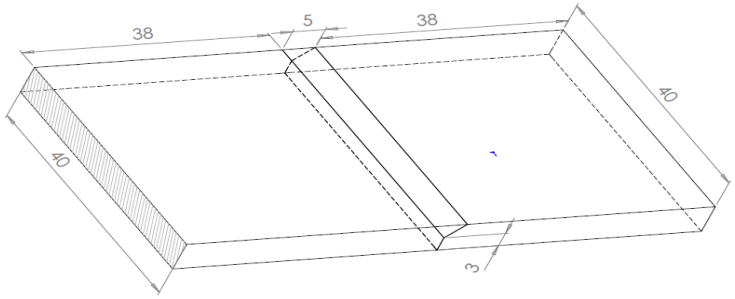
Session: 2023-24
FIRST YEAR SEM-I (Regular)
B. Tech. Branch (AI, CSE-A,CSE-B, DS)
Job Drawing – Welding Shop

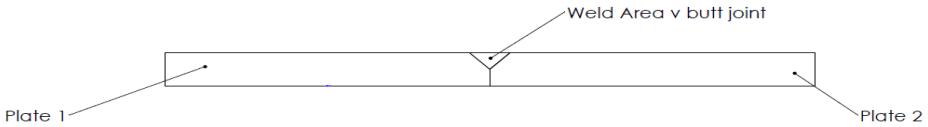
Name of the job: To make Single 'V' Butt Joint.

Require Tools and Equipments: Steel Rule, Try Square, Bench Vice, Flat file, Face Sheild, Flat Tong, Chipping Hammer, Wire Brush, Centre Punch, AC transformer with standard outer accessories.

Raw material: M.S. Flat 40 mm x 40 mm x 05 mm, MS electrode 3.15 mm x 300 mm – 1Nos

Job Drawing:





Note: All Dimensions are in mm **Scale: Not to Scale.**

Job Operations:

Sr. No.	Operation Name	Sr. No.	Operation Name
1	Measuring & Marking	4	Tacking
2	Cutting	5	Welding
3	Filing	6	Cleaning & Finishing

NAME OF THE SHOP :- Welding Shop

NAME OF THE JOB :- To make Single 'V' Butt Joint.

MATERIAL REQUIRED:- M.S. Flat 40 mm x 40 mm x 05 mm—**2 Nos** & MS electrode 3.15 mm x 300 mm – **1 Nos**

TOOLS AND EQPT.REQUIRED:- Steel Rule, Try Square, Bench Vice, Flat file, Face Shield, Flat Tong, Chipping Hammer, Wire Brush, Centre Punch, AC transformer with standard outer accessories.

OPERATION SHEET:-

S.N.	OPERATION	TOOLS REQUIRED	PROCEDURE
1	Measuring & Marking	Steel Rule , pencil	1. Mark length of 40 mm on 50 mm width M.S. flat by pencil.
2	Cutting	Power Hack Saw cutoff machine	1. Cut flat from markings made by pencil. So this way we have to cut two similar flat pieces of dimension(50 x 40 mm) (width x length)
3	Filing	Flat file	1. The Side edges of two pieces are filed such as they are in right angle to each other and width are exactly in given dimensions. 2. Surfaces are filed both the pieces by using the flat file in single V butt joint shape.
4	Tacking	Arc welding machine with standard accessories M.S. Electrode (3.15 mm), Hand Screen	1. Workpieces should be positioned in flat position on the working table. 2. Wear apron & hand gloves. Hold hand screen in hand. 3. Start arc welding machine. Set current amperage 150 amp by rotating current adjustment knob. 4. The arc is strucked and the workpieces are tack welded at the ends of both sides.

5	Welding	Arc welding machine with standard accessories M.S. Electrode (3.15 mm)	1. Welding is then cat thoughout the length of the single 'V' butt joint. 2. Welding run is done with proper weaving and with uniform movement.
6	Finishing & Filing	Flat file, chipping hammer, Wire brush, Tong	1. The scale formation on the welds is removed by using the chipping hammer and wire brush. 2. Filing is done to remove any spatter around the weld.

Result: The Single 'V' Butt Joint is successfully done.